Worksheet 1-6 & 1-7 A Algebraic Limits #1

AP Calculus Name_____

Directions: Evaluate the following limits, if possible. Show all work and use proper notation.

1. $\lim_{x\to 3} \frac{x^2+4x-21}{x-3}$

2. $\lim_{x\to 2} \frac{x^2+3x-10}{x-2}$

3. $\lim_{x\to 2} \frac{e^{7-2x}}{x^3+1}$

4. $\lim_{x\to 1} \frac{x^2-5x+4}{x^2+5x-6}$

5. $\lim_{x\to 0} \frac{x^2}{x^3-3x^2}$

6. $\lim_{x\to \frac{\pi}{3}} \frac{\sin(x)}{x^2}$

7. $\lim_{y\to -2} (2y-3)$

8. $\lim_{x\to -6} \frac{x^2+2x-24}{x^2+x-30}$

9. $\lim_{x\to 0} \frac{x^3-x^2-12x}{x^2-5x}$

10. $\lim_{c\to 3} 5$

11. $\lim_{x \to 5} \frac{x^2 - 10x + 25}{x^2 + x - 30}$

12. $\lim_{x \to 1} \frac{3x^2 - x - 2}{2x^2 + x - 3}$

13. $\lim_{x \to 2} \frac{x^2 + 4x + 7}{x^2 - 3x + 1}$

14. $\lim_{x \to 0} \frac{\sqrt{x + 7} - \sqrt{7}}{x}$

15. $\lim_{x \to 0} \frac{\sqrt{2 + x} - \sqrt{2}}{x}$

16. $\lim_{x \to 4} \frac{\sqrt{x + 5} - 3}{x - 4}$

17. $\lim_{x \to 3} \frac{\sqrt{x + 1} - 2}{x - 3}$

18. $\lim_{x \to -5} \frac{\sqrt{x^2 - 16} - 3}{x + 5}$